



CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

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L24

Sep 21, 2026

Ang and Tang

Extreme Value Distributions **4.2.3**

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Why Study Extreme Value Distinctly?

Fukushima Nuclear Accident (2011)



Do you size the seawall for

(a) the highest flood ever recorded with some margin ($5.7 \text{ m} \times SF$).

OR

(b) something else?

TEPCO
2008: 15.3 m sea height

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Why Study Extreme? The Netherlands

Why Study Extreme? The Netherlands

Netherlands: below,



Canals, Massive Dams, Levees, Dikes

Johan van Veen

1930s - Predicted Dikes are underdesigned

1953s - North sea flood

- Destroying 150+ dikes.
- 1800 people drowned.

Observed maxima

vs.

Expected maxima

Johan van Veen

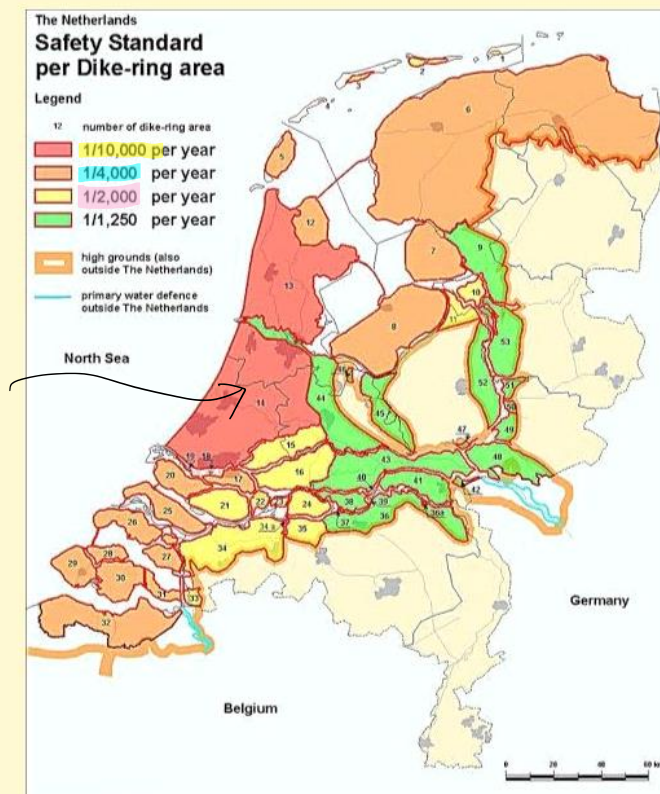


Why Study Extreme? The Netherlands



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Why Study Extreme? The Netherlands



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How to Study Extreme?

How to predict an extreme that has NOT been observed?

Big Idea:

Parent Distribution



Extreme

Distribution

Announcement

- Lectures
 - No class on next Tuesday (29th Sep)
 - Fridays 25th Sep Class at 10 am
- Check GitHub regularly

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Why Study Extreme Value Distinctly?

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Max. Extreme for Exponentially decaying tail

Consider initial variate X to be an exponential RV with the PDF

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

(1) Find the **CDF of maximum extreme** $F_{Y_n}(y)$ of sample size n .

(2) What is the **limiting CDF as $n \rightarrow \infty$** ?

$$Y_n = \max(X_1, X_2, \dots, X_n)$$

$$F_{Y_n}(y) = [F_X(y)]^n$$

$$F_X(x) = \int_0^x f_X(x) dx = 1 - e^{-\lambda x}$$

$$F_{Y_n}(y) = (1 - e^{-\lambda y})^n \quad \text{--- ①}$$

$$= 1 - n e^{-\lambda y} + \frac{n(n-1)}{2!} e^{-2\lambda y} - \frac{n(n-1)(n-2)}{3!} e^{-3\lambda y} + \dots$$

$$\lim_{n \rightarrow \infty} \Rightarrow n(n-1) \approx n^2; \quad n(n-1)(n-2) \approx n^3$$

$$F_{Y_n}(y) \approx \exp[-n e^{-\lambda y}]$$

↑ Double Exponential

Asymptotic Extreme Value Distribution

Type I Dist.
Gumbel Dist.

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Max. Extreme for Polynomially decaying tail

Consider initial variate X to be a Pareto RV with the PDF

$$f_X(x) = \frac{k x_0^k}{x^{k+1}}, \quad x \geq x_0 > 0, k > 0$$

(1) Find the **CDF of maximum extreme** $F_{Y_n}(y)$ of sample size n .

(2) What is the **limiting CDF as $n \rightarrow \infty$** ?

$$F_X(x) = 1 - \left(\frac{x_0}{x}\right)^k$$

$$\text{① } F_{Y_n}(y) = \left[1 - \left(\frac{x_0}{y}\right)^k\right]^n \quad \text{--- EXACT}$$

$$\text{② } \lim_{n \rightarrow \infty}$$

$$F_{Y_n}(y) \approx e^{-n(x_0/y)^k}$$

$$= e^{-(\sqrt[n]{n}/y)^k}$$

$$\text{where } \sqrt[n]{n} = n^{1/k} \cdot x_0$$

Asymptotic ($n \rightarrow \infty$)
single Exponential
Type II EV
Fréchet Dist.

$$= e^{(1/\lambda)} y$$

$$\text{where } \sqrt[n]{n} = n^{1/n} \cdot x_0$$

Fréchet Dist.

Max. Extreme for Exponentially decaying tail

Consider initial variate X to be an exponential RV with the PDF

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

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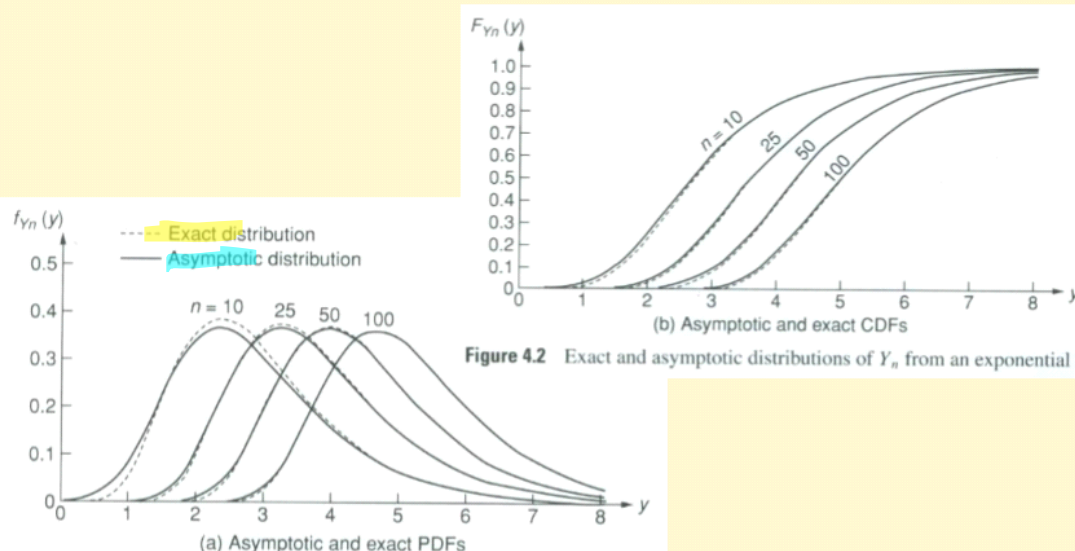
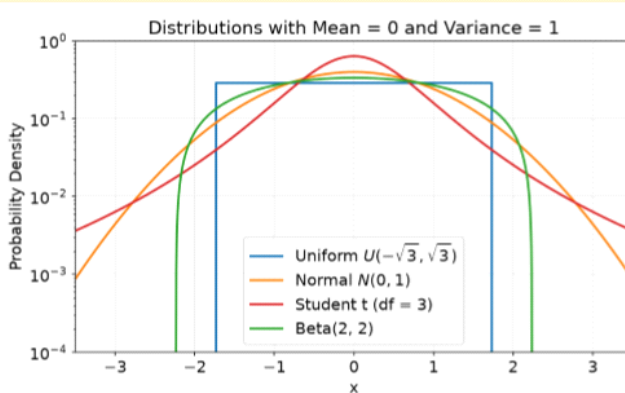
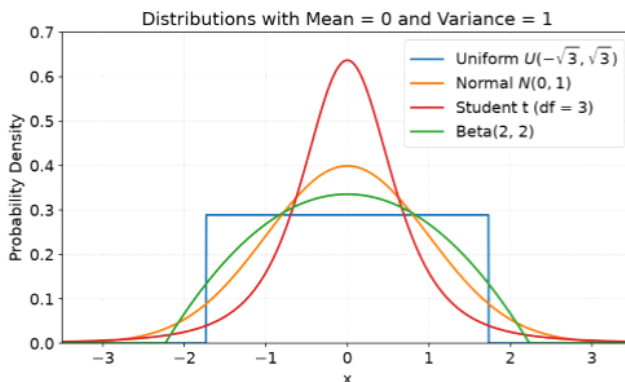


Figure 4.2 Exact and asymptotic distributions of Y_n from an exponential initial distribution.

Recall: dist. tails



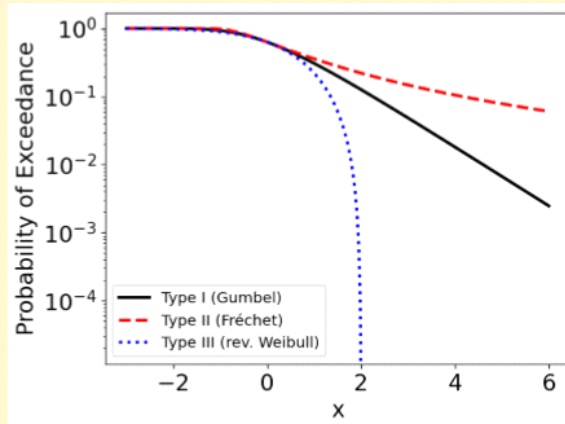
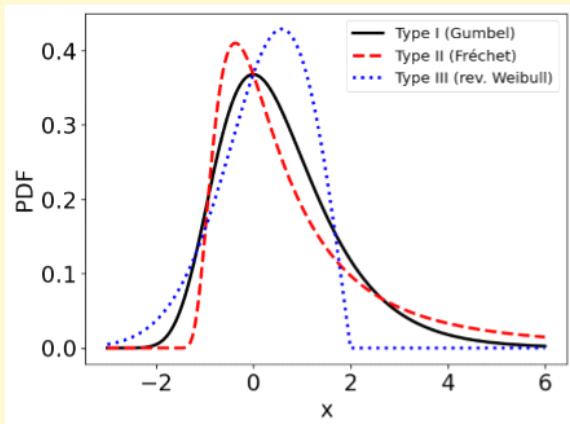
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Forms of Extreme Value Distributions

	Parent Tail Type	Asymptotic EVD
Parent tail	Exponentially decaying tail e.g., Normal, exponential	Double Exp. $F_{Y_n}(y) \sim \exp[-n \exp\{-\lambda y\}]$ Type I; Gumbel
	Polynomially decaying e.g., t-distribution	Single Exp Type II; Fréchet
	Limited (bounded) extreme	Exp with a lower/upper bound Type III; Weibull ↳ Reverse Weibull

Selection of GEV distribution type

Parent distribution	Tail type	Extreme value type



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Extreme value analysis

- Magnitude of extremes
- Frequency of extremes

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Return period

- Four key concepts

- Poisson distribution

Probability of observing x extreme events in T period,

$$\Pr(x, \lambda, T) = \frac{(\lambda T)^x e^{-\lambda T}}{x!} \quad \text{where } x = 0, 1, 2, \dots \text{ and } \lambda > 0$$

$$E[x] = \text{Var}[x] = \lambda T$$

Assumptions: Independent trials; large number of trials; small p

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Return period

- A design event has **10% probability of exceedance in 50 years** ($p_{f,50y} = 0.10$). What is the **annual expected number of events** λ and the **return period** T_r in years?

$$\Pr(x, \lambda, T) = \frac{(\lambda T)^x e^{-\lambda T}}{x!}$$

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Return period

- Given a design life of $T_d = 50$ years and a design event with annual probability ($p_{f,1y} = 1/475$), what is the **probability of failure during the design life, p_{f,T_d}** ?

$$\Pr(x, \lambda, T) = \frac{(\lambda T)^x e^{-\lambda T}}{x!}$$

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Return period

- What is the probability of a 475-year return period event occurring at least once in 475 years?

$$\Pr(x, \lambda, T) = \frac{(\lambda T)^x e^{-\lambda T}}{x!}$$

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Thank you!